

Supplementary Information

Long-period site residuals and response-spectrum hazard sensitivity in Japanese strong-motion records

Supplementary Figures

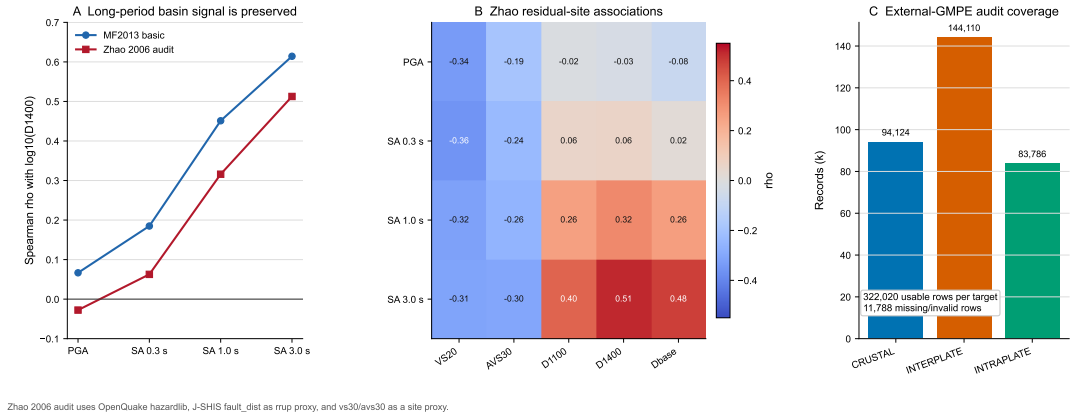


Figure S1. External Zhao et al. OpenQuake sensitivity check. PGA shows weak association with D1400; SA(3.0 s) retains a strong basin-depth association. The check uses distance and site proxy inputs.

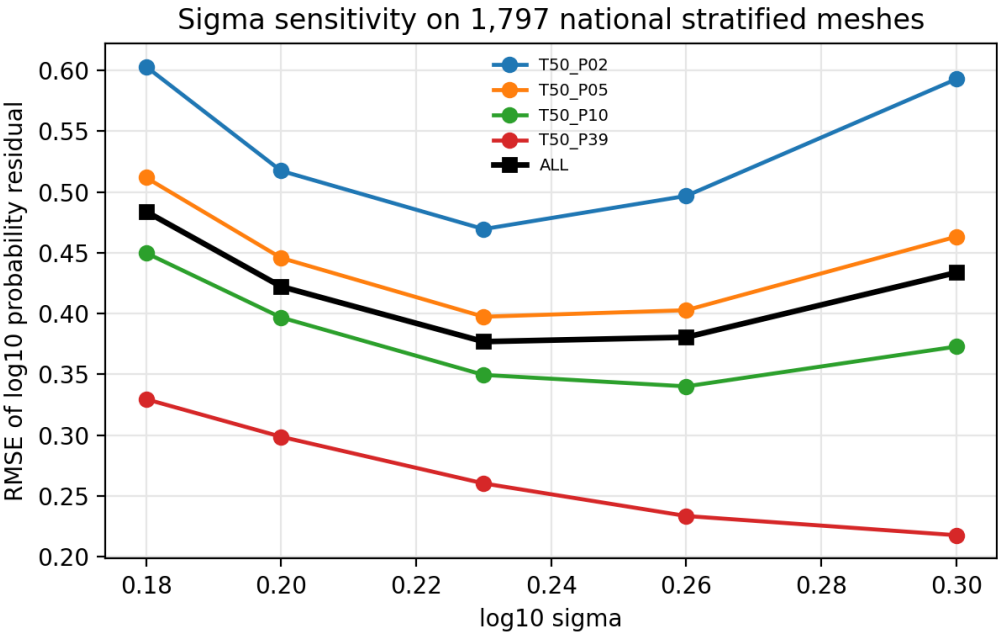


Figure S5. Five-point sigma sensitivity audit for the public-parameter J-SHIS PBV recomputation. The same 1,797 national stratified meshes are evaluated at log10 sigma values of 0.18, 0.20, 0.23, 0.26, and 0.30. The global RMSE minimum occurs at sigma 0.23; probability-level optima vary across exceedance levels.

Ablation, interpretability, transfer, and engineering scale of the station-residual model

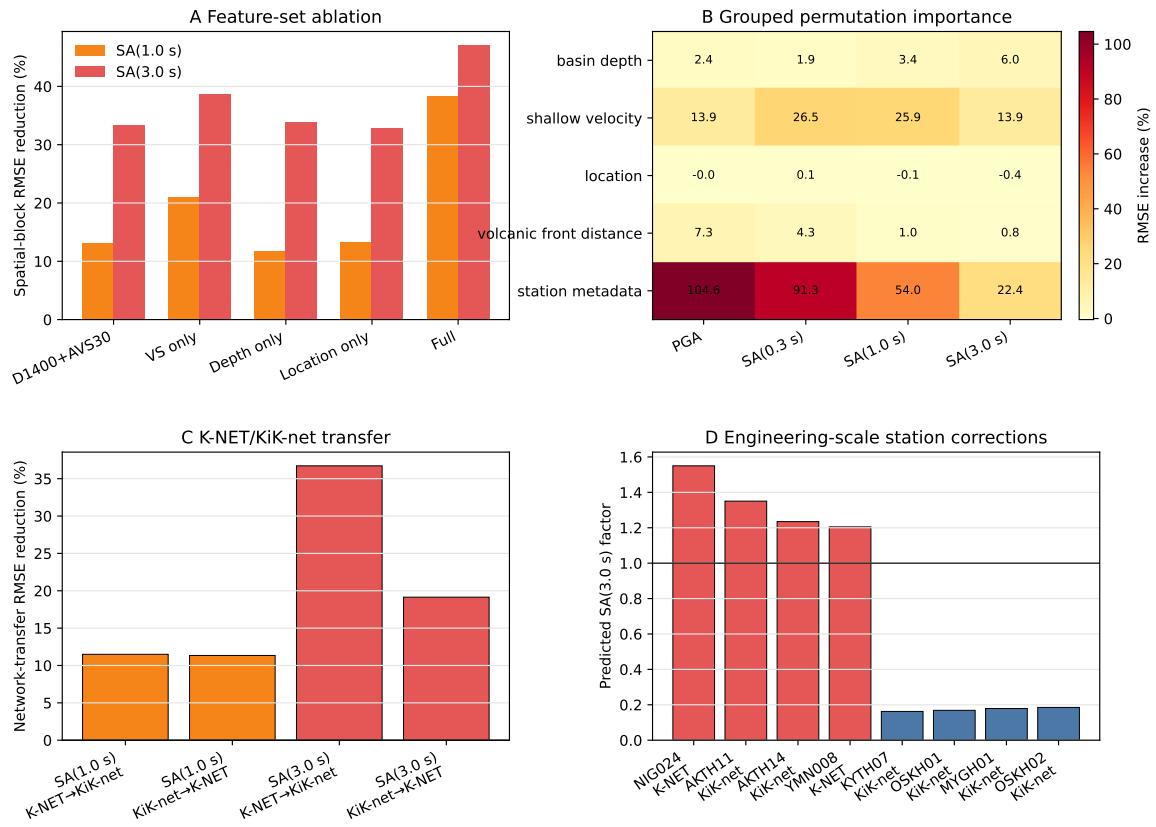


Figure S2. Ablation, interpretability, network-transfer validation, and engineering-scale examples for the station-residual model. A: spatial-block feature-set ablation for SA(1.0 s) and SA(3.0 s). B: grouped permutation importance. C: K-NET/KiK-net transfer. D: held-out SA(3.0 s) station correction factors.

Spatial stability and uncertainty of official SA(3.0 s) response-spectrum check

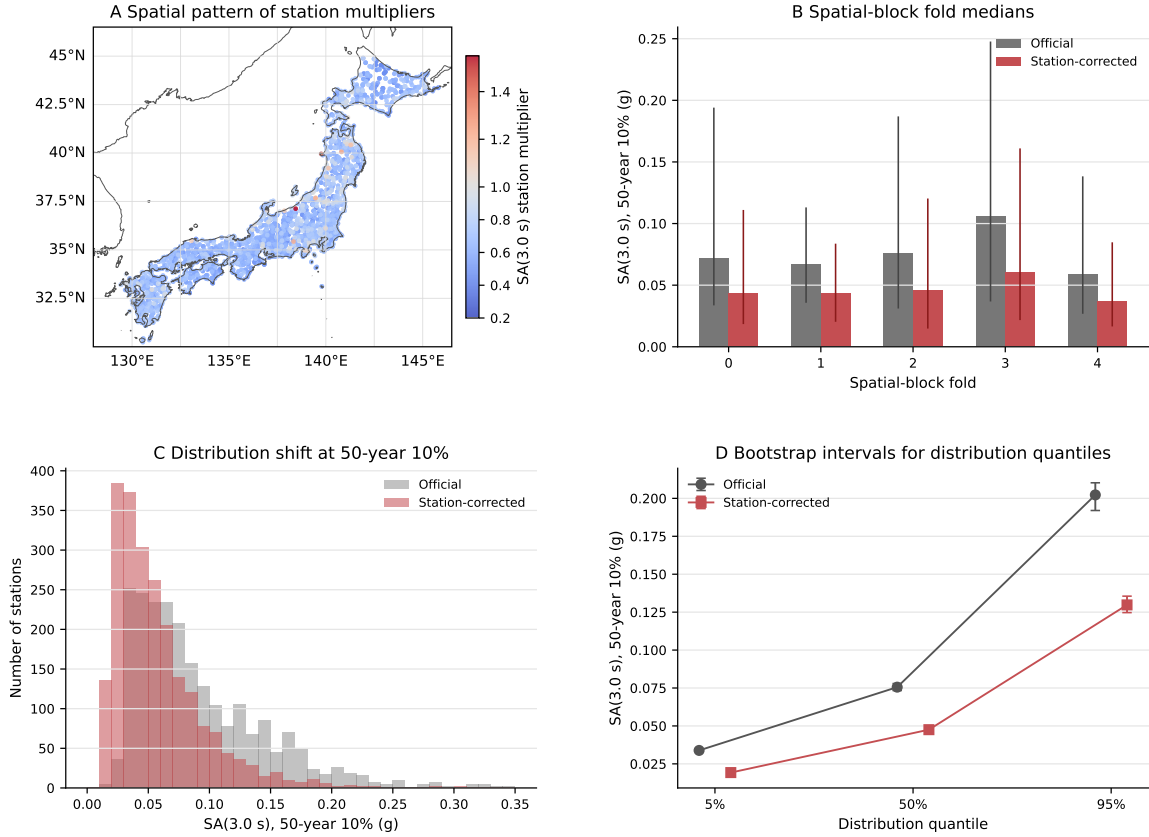


Figure S3. Spatial stability and uncertainty of the official SA(3.0 s) response-spectrum check. A: station multiplier map for matched held-out stations. B: spatial-block fold medians, with vertical lines showing 5th–95th percentile ranges. C: distribution shift from official to station-corrected 50-year 10% SA(3.0 s). D: station-bootstrap 95% intervals for the 5th, 50th, and 95th percentiles.

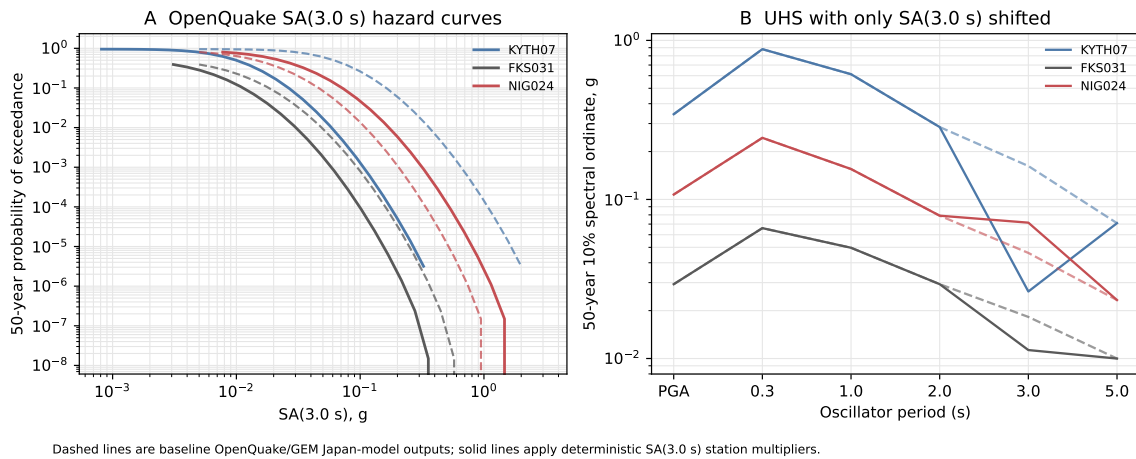


Figure S4. Independent OpenQuake/GEM Japan-model PSHA sensitivity check for representative stations. A: mean SA(3.0 s) hazard curves from the OpenQuake baseline calculation, with dashed lines showing baseline curves and solid lines showing deterministic SA(3.0 s) station-multiplier shifts. B: 50-year 10% uniform-hazard spectra for the same representative stations, with only the SA(3.0 s) ordinate shifted. This supplementary calculation provides an engine-level sensitivity check using packaged OpenQuake Japan mosaic inputs.

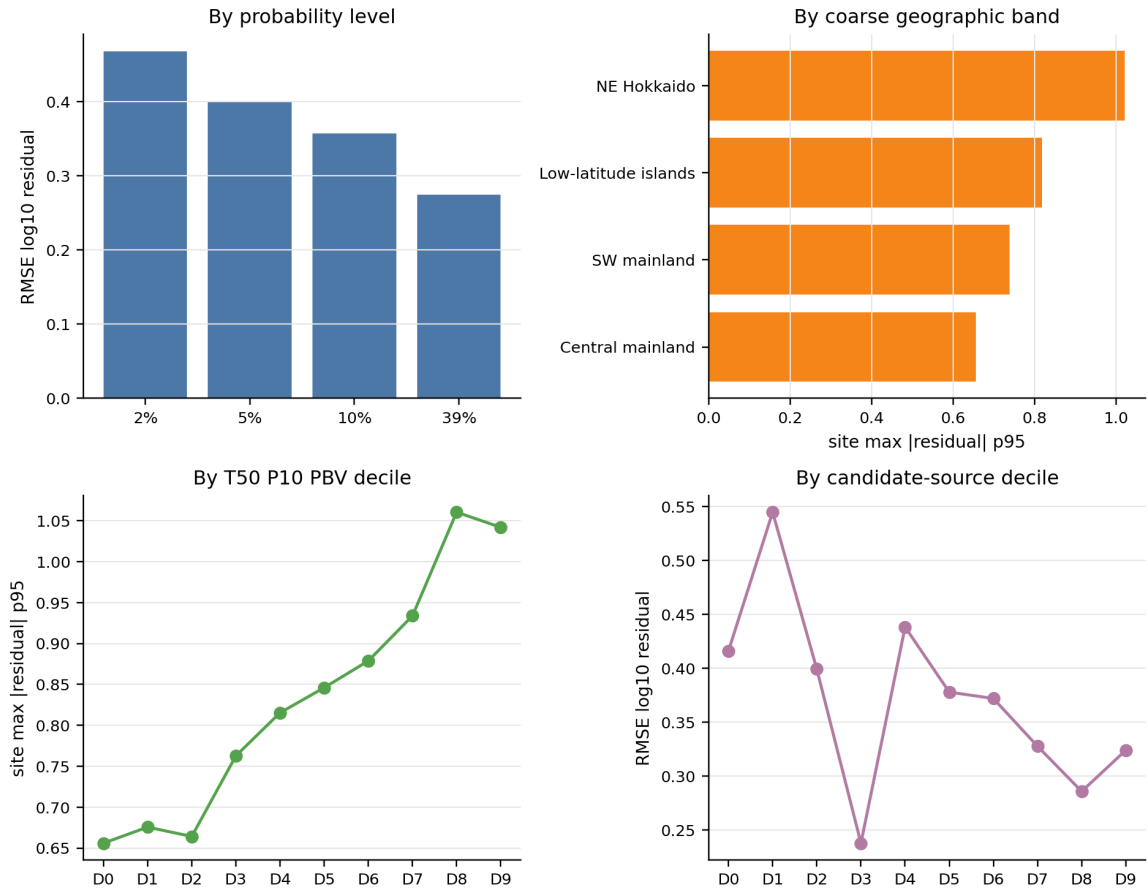


Figure S6. Residual-tail diagnostics for the 5,322-mesh public-parameter audit. A: RMSE by official probability level. B: site-level tail amplitude by coarse geographic band. C: site-level tail amplitude by official PBV decile. D: RMSE by candidate-source-count decile. These diagnostics identify structured public-parameter mismatch and confirm valid numerical execution.

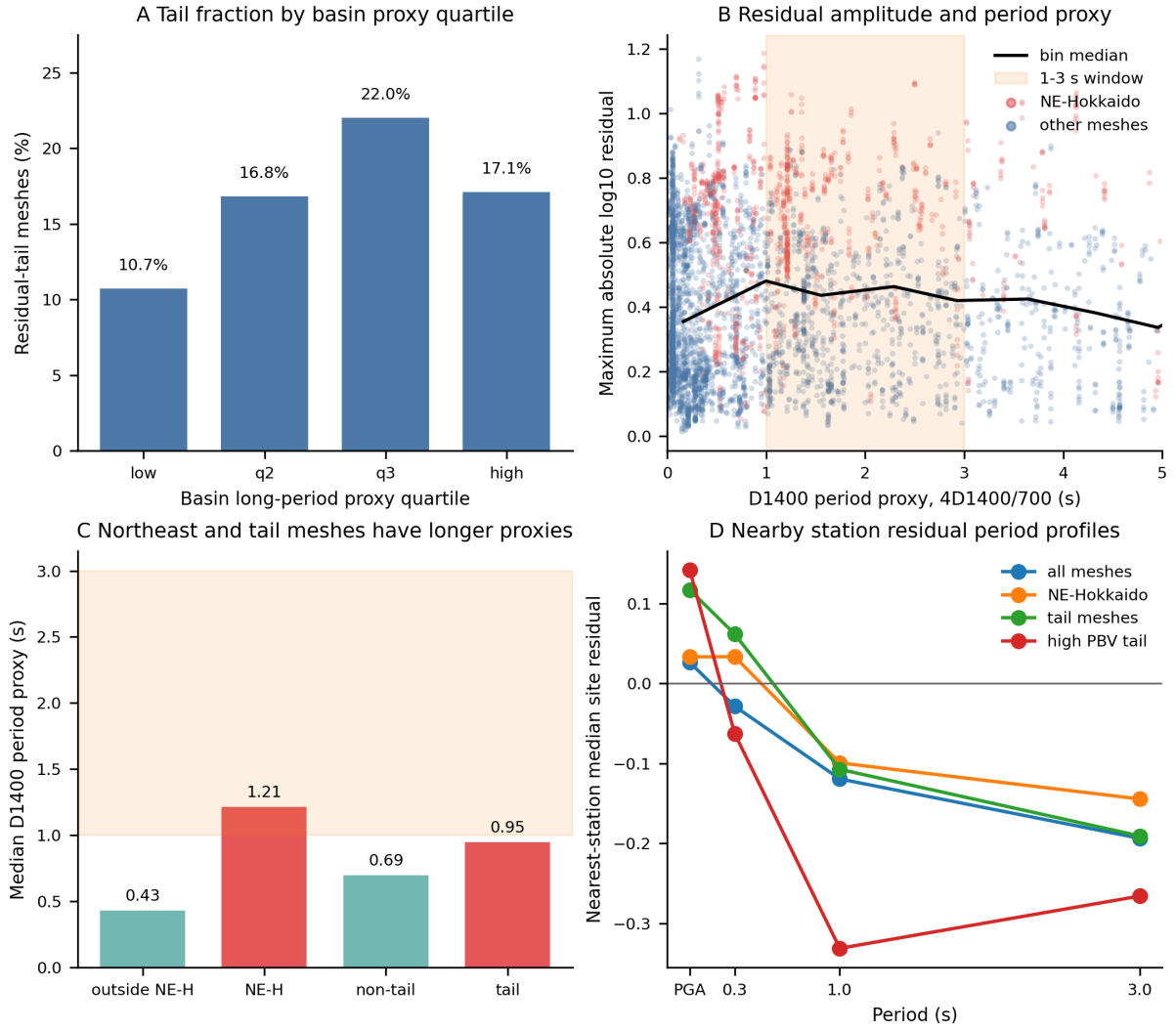


Figure S7. Basin-period proxy audit for the 10,467-mesh public-parameter residual-tail structure. A: residual-tail fraction by basin long-period proxy quartile. B: maximum absolute log10 residual against the D1400-derived period proxy $4D1400/700$, with the 1–3 s window shaded. C: median D1400-derived period proxy for northeast-Hokkaido, outside-northeast-Hokkaido, residual-tail, and non-tail meshes. D: nearby-station MF2013 site-corrected residual period profiles for all meshes, northeast-Hokkaido meshes, residual-tail meshes, and high-PBV tail meshes.

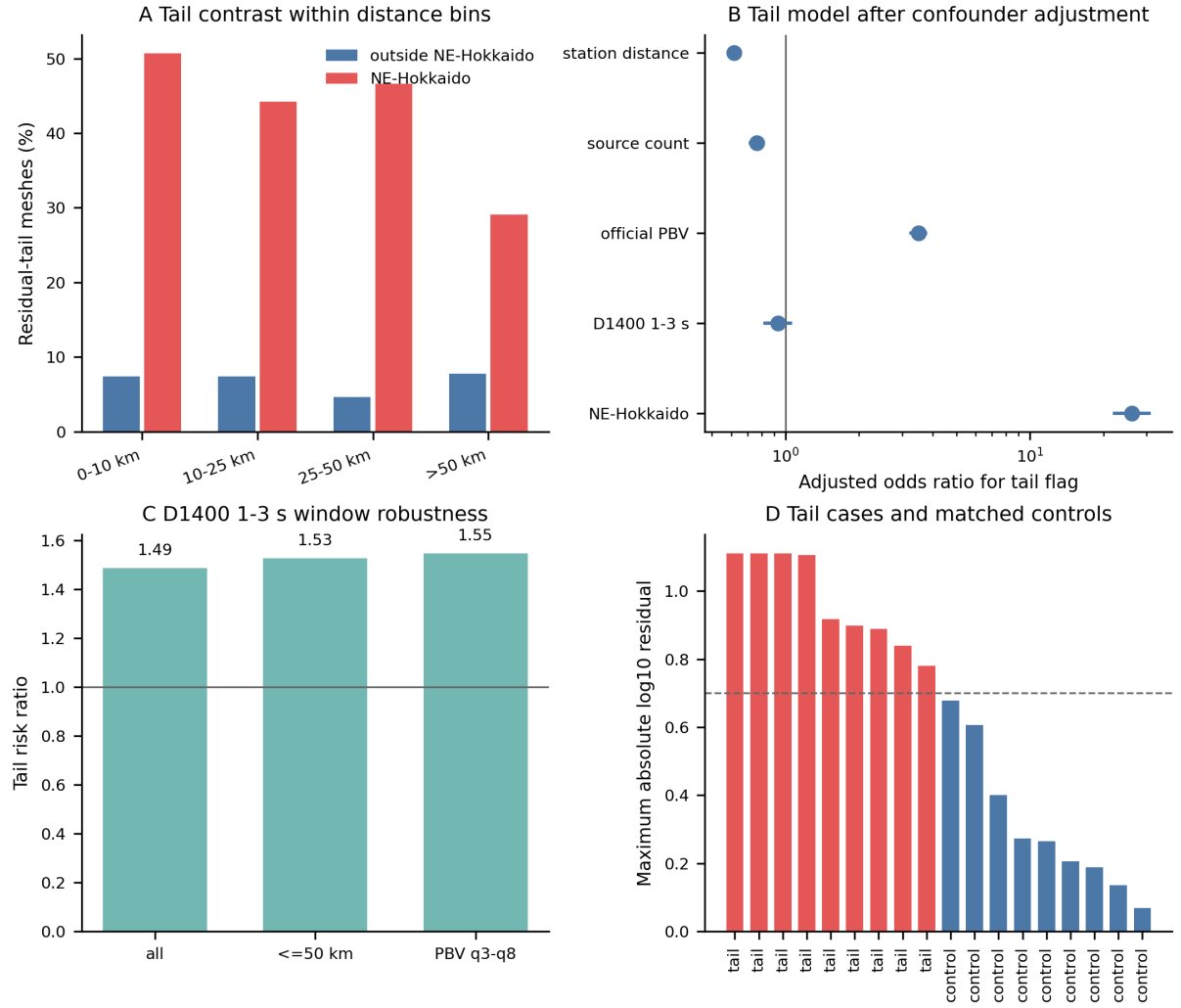


Figure S8. Tail-robustness and matched-case audit for the 10,467-mesh public-parameter residual-tail structure. A: residual-tail fraction in northeast-Hokkaido and outside-northeast-Hokkaido meshes within station-distance bins. B: adjusted odds ratios from a logistic tail model with official PBV, candidate-source count, station distance, station residual contrast, northeast-Hokkaido membership, and the D1400-derived 1–3 s proxy window. C: tail-risk ratios for the D1400-derived 1–3 s proxy window across all meshes, the nearest-station ≤ 50 km subset, and middle official-PBV deciles q3–q8. D: maximum absolute log10 residuals for matched tail and non-tail mesh profiles.

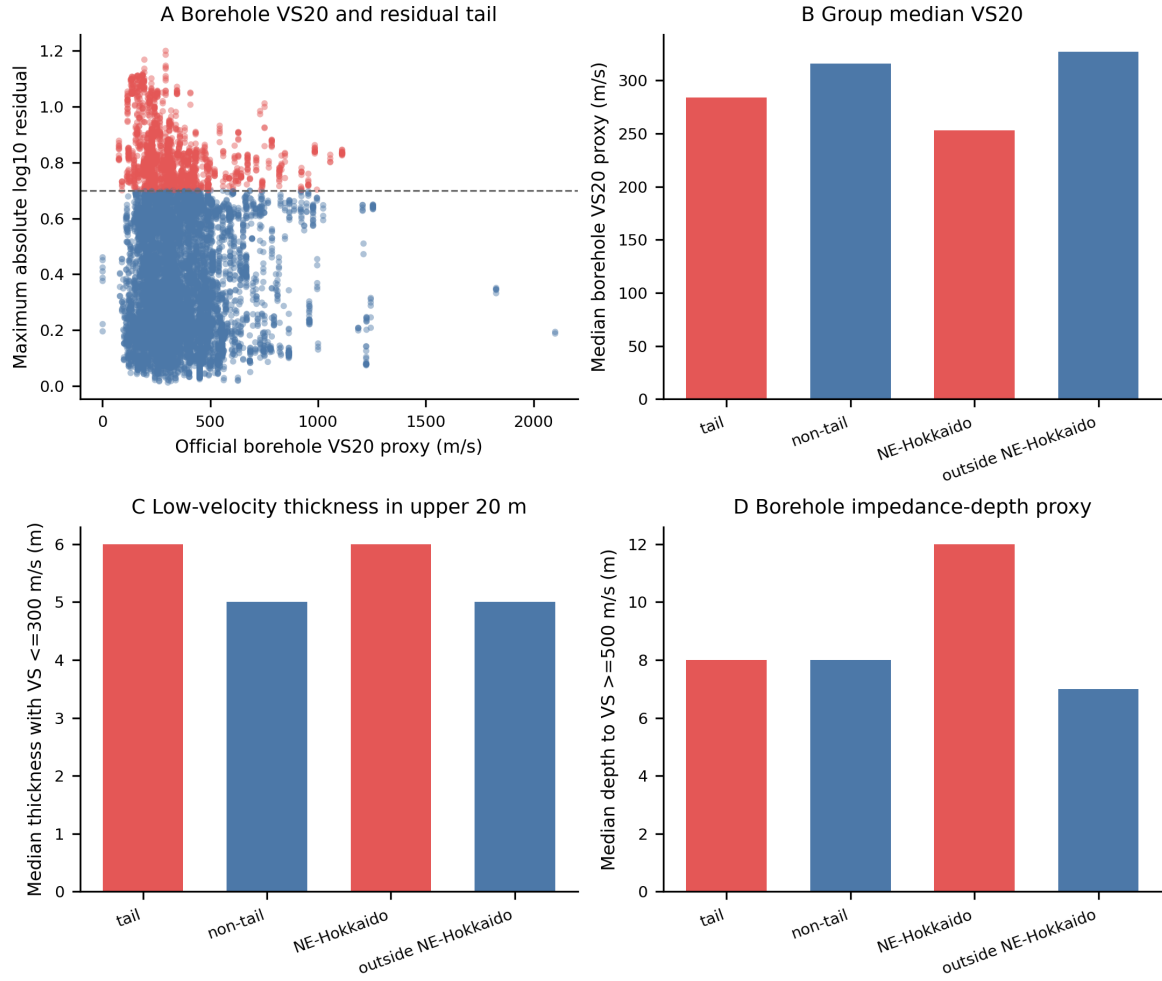


Figure S9. Nearest-station official borehole-profile audit for the 10,467-mesh public-parameter residual-tail structure. A: maximum absolute log10 residual against the public NIED borehole VS20 proxy for the nearest K-NET/KiK-net station, with residual-tail meshes highlighted. B: median borehole VS20 proxy for tail, non-tail, northeast-Hokkaido, and outside-northeast-Hokkaido groups. C: median upper-20 m thickness with $VS \leq 300$ m/s. D: median depth to $VS \geq 500$ m/s. The audit provides a nearest-station profile check for proxy-level interpretation.

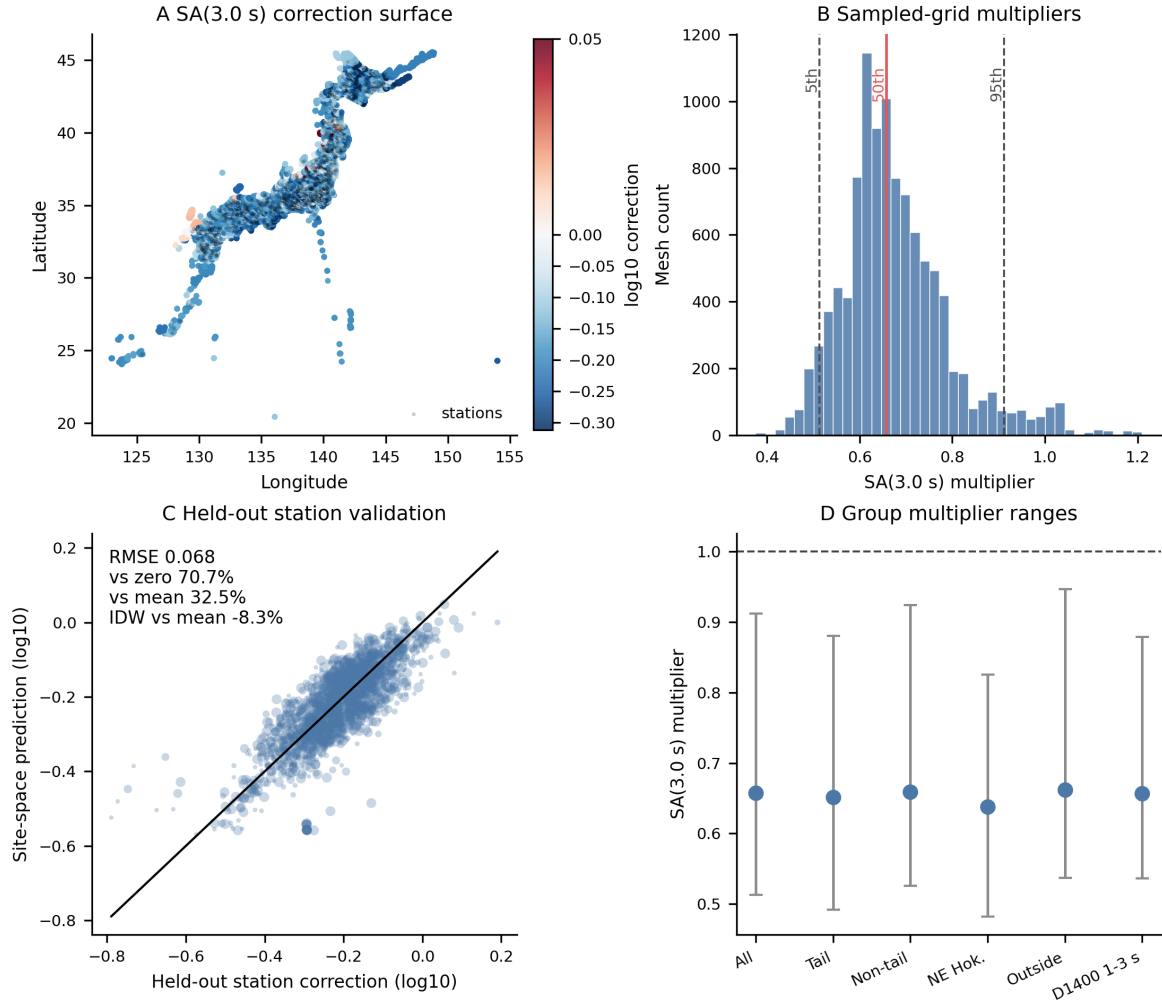


Figure S10. Sampled-grid SA(3.0 s) continuous correction-surface audit. A: site-and-location gradient-boosted correction surface on the 10,467-mesh public-parameter audit grid, with station locations shown for reference. B: sampled-grid multiplier distribution. C: spatial-block held-out station validation for the site-and-location surface. D: group median multipliers with 5th–95th percentile ranges. The sampled-grid values use nearest-station public site proxies and test the station-to-grid continuity step.

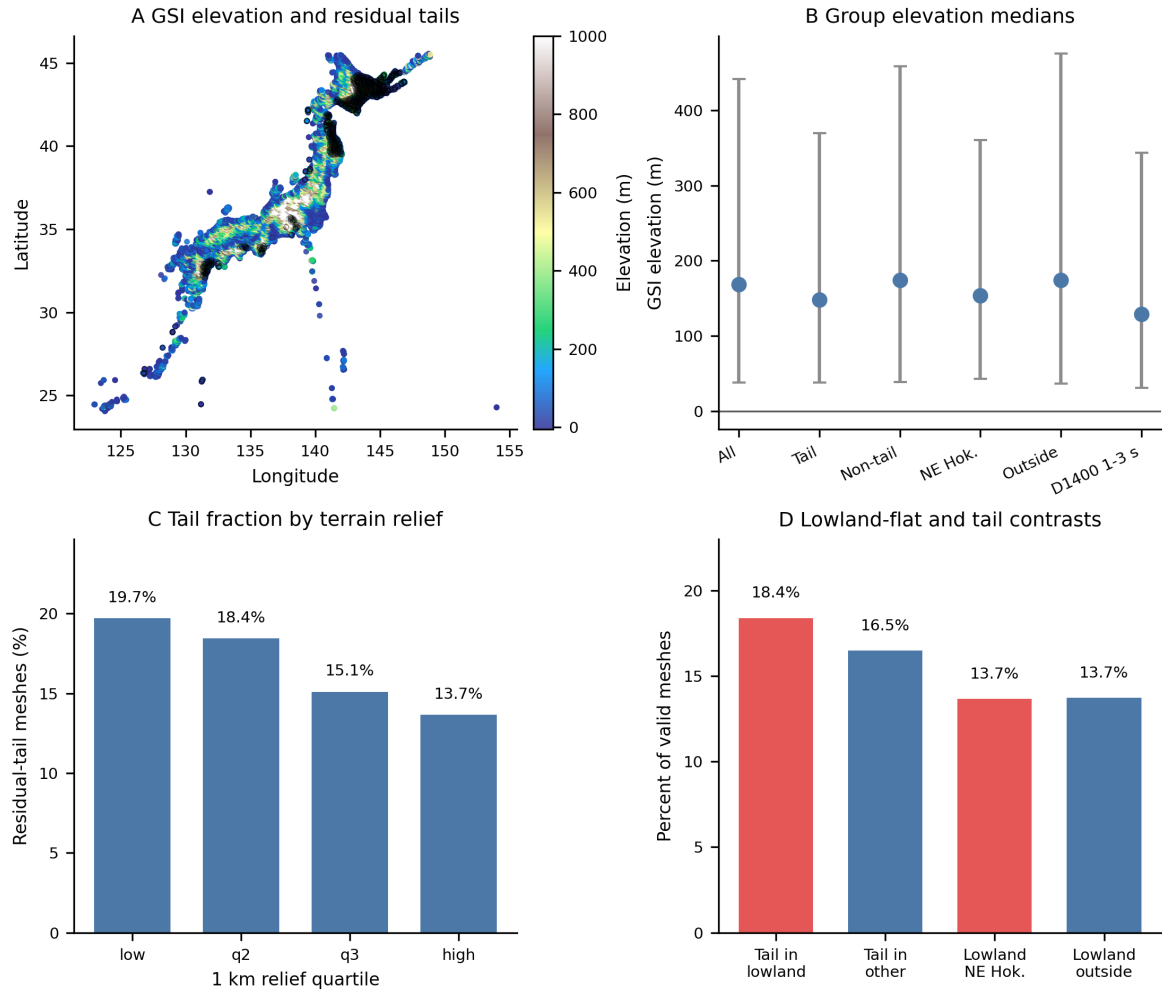


Figure S11. Independent GSI terrain geomorphic proxy audit for the 10,467-mesh public-parameter residual-tail structure. A: GSI elevation sampled from public DEM PNG tiles, with residual-tail meshes outlined. B: group elevation medians with interquartile ranges. C: residual-tail fraction by 1 km relief quartile. D: lowland-flat terrain and residual-tail contrasts. The audit tests external surface-geomorphic consistency; shear-wave velocity and three-dimensional basin geometry require separate data.

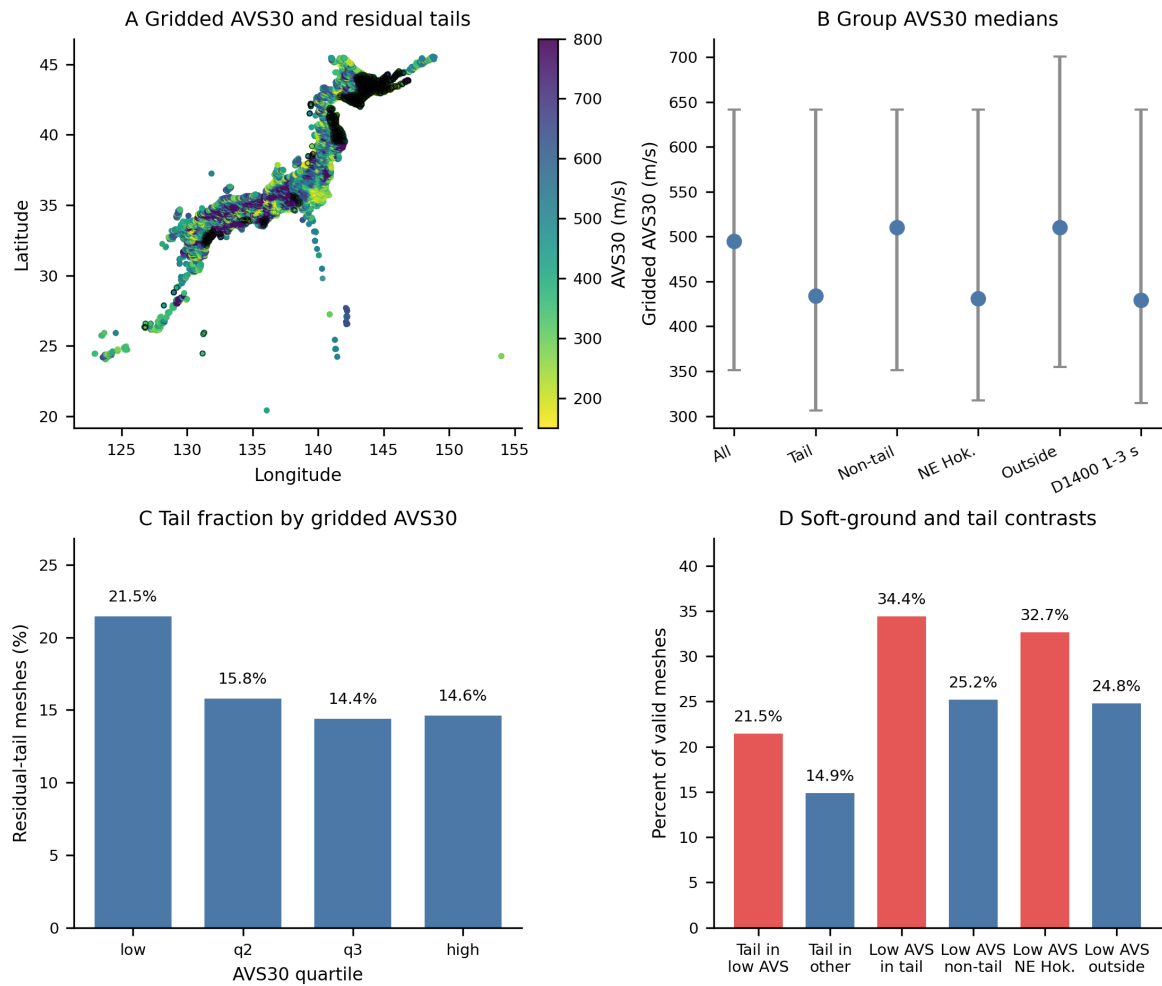


Figure S12. J-SHIS VS400 site-amplification layer audit for the 10,467-mesh public-parameter residual-tail structure. A: gridded AVS30 from the public 250 m surface-ground layer, with residual-tail meshes outlined. B: group AVS30 medians with interquartile ranges. C: residual-tail fraction by gridded AVS30 quartile. D: soft-ground and residual-tail contrasts. The audit supplies gridded shallow-velocity and surface-amplification context for the residual-tail interpretation.

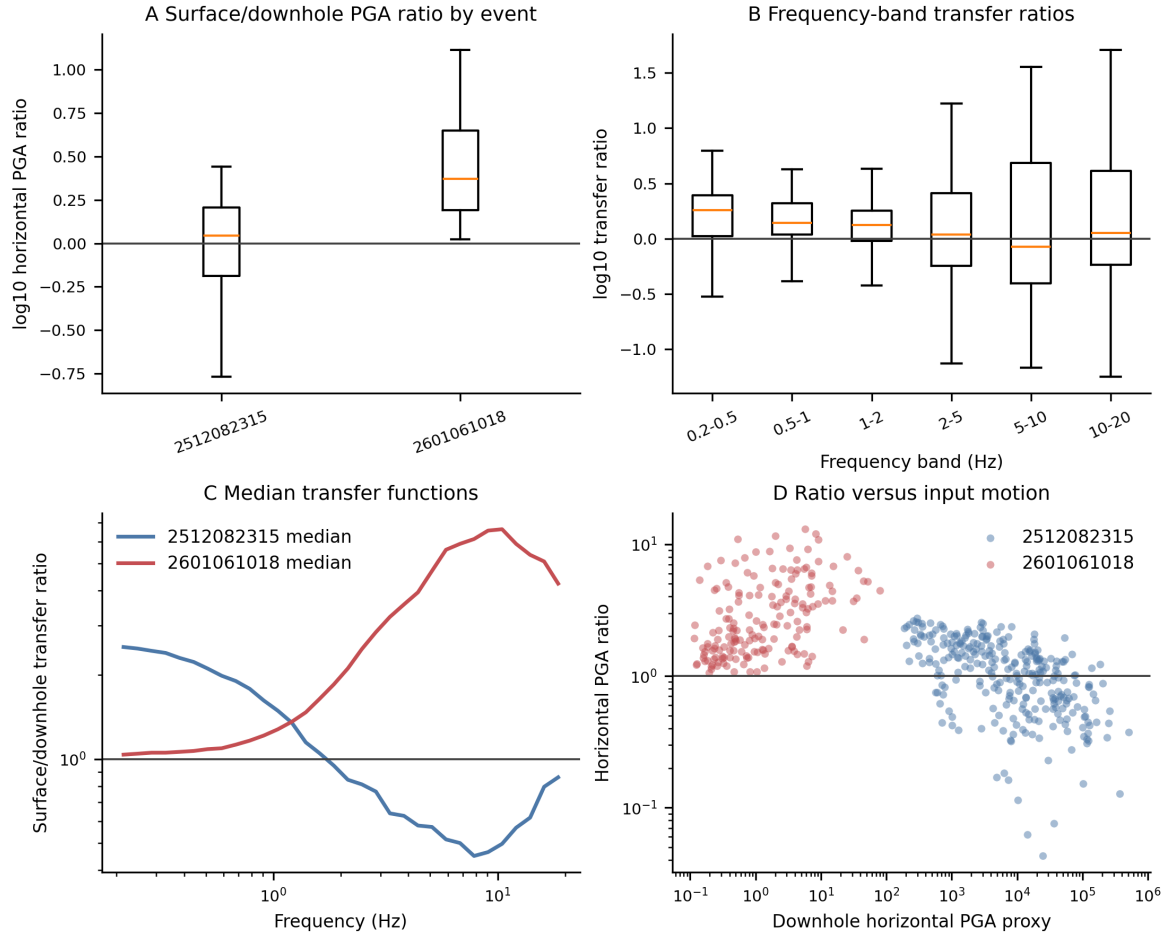


Figure S13. Multi-event KiK-net surface/downhole transfer-function audit. A: horizontal PGA ratio by event. B: frequency-band transfer-ratio distributions for 460 event-station pairs. C: event-wise median transfer functions. D: horizontal PGA ratio against downhole input-motion proxy. The audit combines 288 converted miniSEED surface/downhole pairs and 172 NIED ASCII pairs converted with header scale factors.

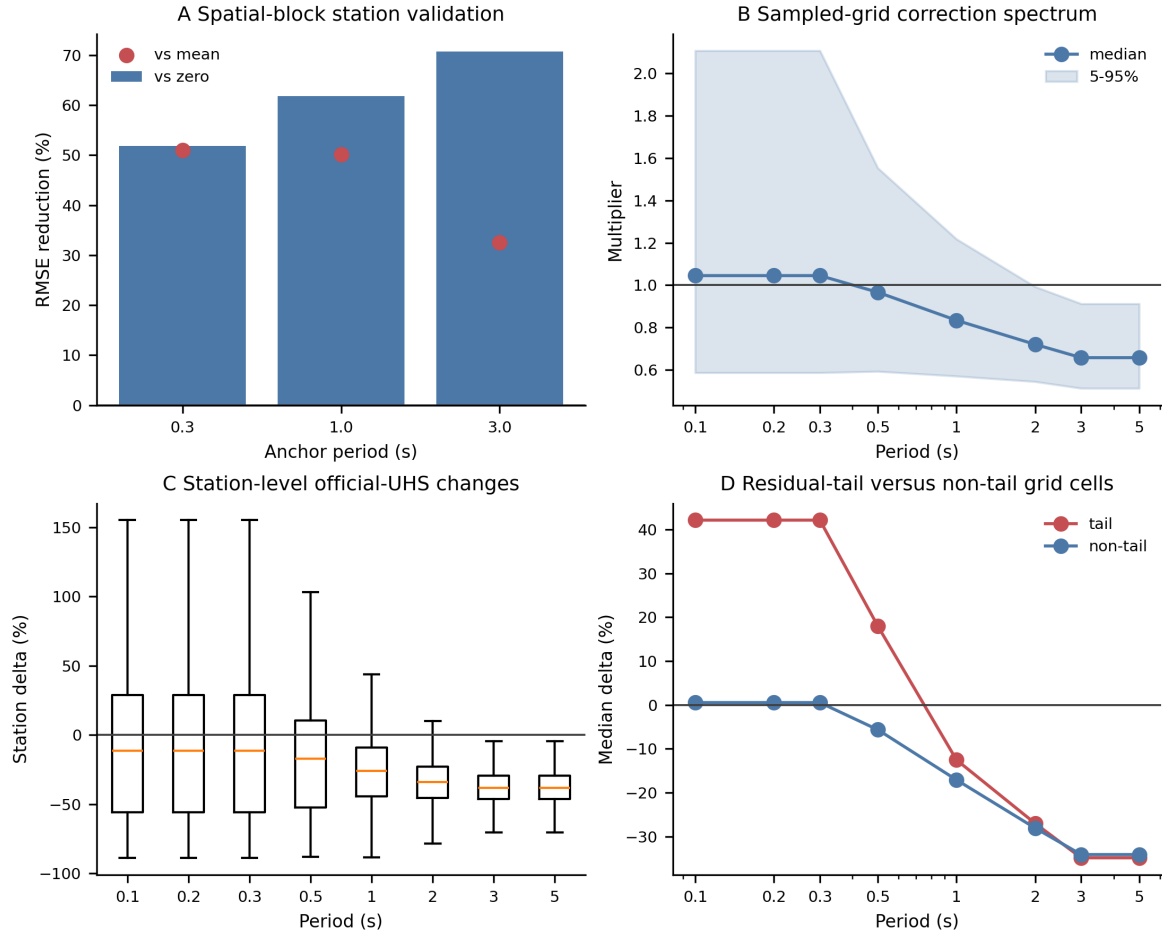


Figure S14. All-period official-response-spectrum non-ergodic propagation audit. A: spatial-block held-out station validation for SA(0.3 s), SA(1.0 s), and SA(3.0 s) correction surfaces. B: sampled-grid 50-year 10% multiplier spectrum on 10,467 meshes. C: matched-station official-UHS percent changes. D: residual-tail and non-tail sampled-grid median changes by period. The calculation propagates station terms through official J-SHIS response-spectrum map ordinates and keeps the official source model and uncertainty settings fixed.

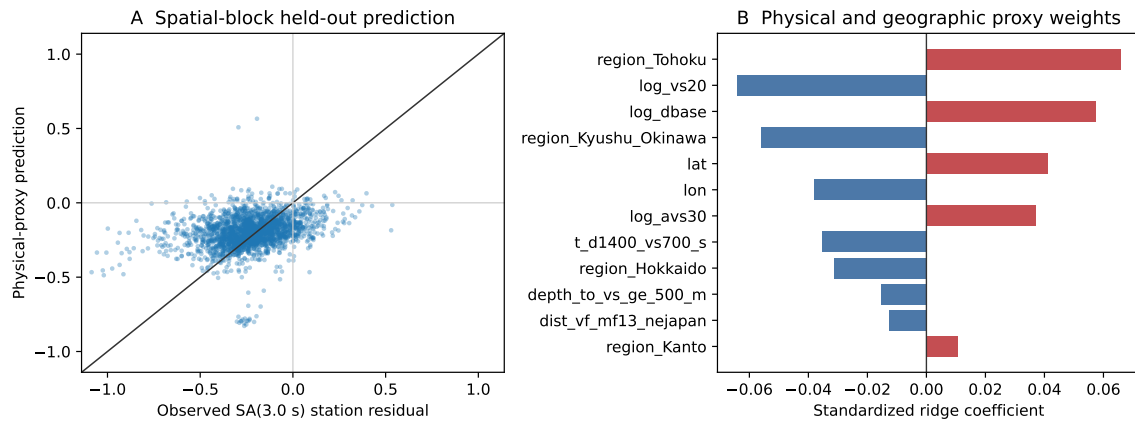


Figure S15. Interpretable SA(3.0 s) site-variable model. A: spatial-block held-out observed and predicted MF2013 site-corrected station residuals. B: largest standardized ridge-regression coefficients. Predictors include basin-depth proxies, shallow-velocity proxies, public borehole-profile proxies, volcanic-front distances, elevation, sensor depth, coordinates, and broad Japanese region.

Supplementary Tables and Audit Files

The accompanying `supplement/` directory contains machine-readable CSV and Markdown audit files for the additional validation checks.

- Supplementary Table S1: `jshis_station_matching_missingness_audit.csv`. This file documents 333,808 public flatfile records, 2,581 record station identifiers, 2,607 site-schema rows, complete record-to-site matching, and site-variable availability.
- Supplementary Table S2: `jshis_station_region_leaveout_summary.csv` and `jshis_station_region_leaveout_fold_metrics.csv`. These files report seven-region geographic leave-out validation for the station-residual model.
- Supplementary Table S3: `jshis_station_variable_substitution_summary.csv`, `jshis_station_site_variable_collinearity_pairs.csv`, and `jshis_station_site_variable_vif.csv`. These files report official-variable substitutions and collinearity checks for D1400, AVS30, VS20, and Dbase.
- Supplementary Table S4: `jshis_official_hazard_response_summary.csv`. This file reports official and SA(3.0 s)-corrected response-spectrum ordinates at 50-year 2%, 5%, 10%, and 39% exceedance probabilities.
- Supplementary Table S5: `jshis_positive_sa3_station_correction_summary.csv` and `jshis_positive_sa3_station_correction_audit.csv`. These files identify the 24 matched stations with positive SA(3.0 s) correction factors and list their site-level values.
- Supplementary Table S6: `jshis_official_data_provenance_2026-06-14.csv`, `cee_code_data_release_manifest_2026-06-14.md`, and `cee_release_metadata_zenodo_draft.json`. These files provide data-source provenance, redistribution boundaries, and DOI-deposition metadata for release preparation.
- Supplementary Table S7: `jshis_existing_sensitivity_inventory.csv` and `jshis_existing_influence_and_external_model_sensitivity.md`. These files inventory the existing event-exclusion, station-exclusion, magnitude-bin, distance-bin, and Zhao et al. external-model sensitivity outputs.
- Supplementary Table S8: `openquake_jpn_station_psha_summary.csv`, `openquake_jpn_station_psha_uhs.csv`, `openquake_jpn_station_psha_hazard_curves.csv`, and `openquake_jpn_station_psha_sensitivity.md`. These files report the independent OpenQuake/GEM Japan-model PSHA sensitivity calculation for five representative stations.
- Supplementary Table S9: `jshis_public_psha_national_stratified5322_arrays12_recompute_audit.md` and `jshis_public_psha_national_stratified5322_arrays12_run_summary.csv`. These files report the 5,322-mesh national stratified public-parameter audit.
- Supplementary Table S10: `jshis_public_psha_sigma_sensitivity1797_audit.md`, `jshis_public_psha_sigma_sensitivity1797_metrics.csv`, and `jshis_public_psha_sigma_sensitivity1797_best.csv`. These files report the five-point sigma sensitivity audit.
- Supplementary Table S11: `jshis_public_psha_residual_root_cause5322_audit.md`, `jshis_public_psha_residual_root_cause5322_by_probability.csv`, `jshis_public_psha_residual_root_cause5322_by_coarse_geo.csv`, `jshis_public_psha_residual_root_cause5322_by_official_bv_quantile.csv`, `jshis_public_psha_residual_root_cause5322_by_candidate_quantile.csv`, and `jshis_public_psha_residual_root_cause5322_spearman.csv`. These files report residual-tail diagnostics for the 5,322-mesh audit.
- Supplementary Table S12: `jshis_public_psha_national_stratified10467_arrays8_recompute_audit.md`, `jshis_public_psha_national_stratified10467_arrays8_recompute_run_summary.csv`, `jshis_public_psha_national_stratified10467_arrays8_recompute_by_probability.csv`, `jshis_public_psha_national_stratified10467_arrays8_recompute_by_coarse_geo.csv`, `jshis_public_psha_national_stratified10467_arrays8_recompute_by_official_bv_quantile.csv`, `jshis_public_psha_national_stratified10467_arrays8_recompute_by_candidate_quantile.csv`, and `jshis_public_psha_national_stratified10467_arrays8_recompute_spearman.csv`. These files report the 10,467-mesh national stratified public-parameter execution and residual-stratification audit.
- Supplementary Table S13: `jshis_public_psha_10467_near_station_physical_audit.md`, `jshis_public_psha_10467_near_station_physical_summary.csv`, and `jshis_public_psha_10467_near_station_physical_link.csv`. These files link the 10,467 sampled meshes to nearby K-NET/KiK-net stations with SA(3.0 s) site residuals and official site fields, and summarize the basin-proxy contrasts for residual-tail and northeast-Hokkaido meshes.
- Supplementary Table S14: `jshis_public_psha_10467_basin_period_proxy_audit.md`, `jshis_public_psha_10467_basin_period_proxy_summary.csv`, `jshis_public_psha_10467_basin_period_proxy_associations.csv`, `jshis_public_psha_10467_basin_period_proxy_tail_risk.csv`, and `jshis_public_psha_10467_basin_period_proxy_link.csv`. These files report the D1400-derived basin-period proxy audit and mesh-level links used for Supplementary Fig. S7.
- Supplementary Table S15: `jshis_public_psha_10467_tail_robustness_cases_audit.md`, `jshis_public_psha_10467_tail_robustness_distance_bins.csv`, `jshis_public_psha_10467_tail_robustness_stratified.csv`, `jshis_public_psha_10467_tail_threshold_sensitivity.csv`, `jshis_public_psha_10467_tail_robustness_logistic.csv`, `jshis_public_psha_10467_tail_case_profiles.csv`, and `jshis_public_psha_10467_tail_case_probability_rows.csv`. These files report the tail-robustness stratification, tail-threshold sensitivity, adjusted logistic tail model, matched tail/non-tail case profiles, and probability-level residual rows used for Supplementary Fig. S8.
- Supplementary Table S16: `jshis_public_psha_10467_nearest_station_borehole_audit.md`, `jshis_public_psha_10467_nearest_station_borehole_summary.csv`, and `jshis_public_psha_10467_nearest_station_borehole_link.csv`. These files report the nearest station borehole audit and mesh-level links used for Supplementary Fig. S9.

- psha_10467_nearest_station_borehole_summary.csv, jshis_public_psha_10467_nearest_station_borehole_profiles.csv, and jshis_public_psha_10467_nearest_station_borehole_mesh_link.csv. These files report the public NIED nearest-station borehole-profile audit used for Supplementary Fig. S9.
- Supplementary Table S17: jshis_sa3_continuous_correction_surface_audit.md, jshis_sa3_continuous_correction_surface_summary.csv, jshis_sa3_continuous_correction_surface_station_validation.csv, and jshis_sa3_continuous_correction_surface_mesh.csv. These files report the sampled-grid SA(3.0 s) continuous correction-surface audit used for Supplementary Fig. S10.
 - Supplementary Table S18: jshis_gsi_terrain_geomorphic_audit.md, jshis_gsi_terrain_geomorphic_summary.csv, jshis_gsi_terrain_geomorphic_associations.csv, and jshis_gsi_terrain_geomorphic_mesh.csv. These files report the independent GSI terrain geomorphic proxy audit used for Supplementary Fig. S11.
 - Supplementary Table S19: jshis_zamp_vs400_site_amplification_audit.md, jshis_zamp_vs400_site_amplification_summary.csv, jshis_zamp_vs400_site_amplification_associations.csv, jshis_zamp_vs400_site_amplification_jcode.csv, and jshis_zamp_vs400_site_amplification_mesh.csv. These files report the public J-SHIS VS400 site-amplification audit used for Supplementary Fig. S12.
 - Supplementary Table S20: cee_claim_evidence_traceability.md and cee_claim_evidence_traceability.csv. These files map the manuscript's main claims to the figures, derived tables, audit files, numerical anchors, interpretation boundaries, and release status that support them.
 - Supplementary Table S21: kiknet_surface_downhole_audit.md, kiknet_surface_downhole_ratios.csv, and kiknet_surface_downhole_audit.py. These files report the earlier paired KiK-net surface/downhole waveform proxy audit retained for comparison with the expanded transfer-function audit.
 - Supplementary Table S22: kiknet_multievent_transfer_function_audit.md, kiknet_multievent_transfer_functions.csv, kiknet_multievent_transfer_function_bins.csv, kiknet_multievent_transfer_function_event_summary.csv, kiknet_multievent_transfer_function_station_summary.csv, and kiknet_multievent_transfer_function_audit.py. These files report and reproduce the two-event KiK-net transfer-function audit used in Fig. 8A and Supplementary Fig. S13.
 - Supplementary Table S23: jshis_multiperiod_nonergodic_psha_audit.md, jshis_multiperiod_station_corrections.csv, jshis_multiperiod_station_surface_validation.csv, jshis_multiperiod_station_surface_validation_summary.csv, jshis_multiperiod_continuous_correction_surface_mesh.csv, jshis_multiperiod_official_response_station_values.csv, jshis_multiperiod_official_response_mesh_values.csv, jshis_multiperiod_nonergodic_psha_summary.csv, and jshis_multiperiod_nonergodic_psha_audit.py. These files report and reproduce the all-period official-response-spectrum propagation audit used in Supplementary Fig. S14.
 - Supplementary Table S24: jshis_official_psha_input_chain_audit.md, jshis_official_psha_input_audit_summary.csv, jshis_official_psha_input_file_inventory.csv, jshis_official_psha_input_shape_inventory.csv, jshis_official_psha_input_download_manifest.csv, and jshis_official_psha_input_audit.py. These files document and reproduce the official J-SHIS 2024 PSHM input-chain audit used to define the source-model boundary for the official-map propagation calculation.
 - Supplementary Table S25: jshis_representative_city_spectrum_cases.csv, jshis_representative_uncertainty_summary.csv, and build_figure9_en.py. These files document and reproduce the representative city-nearest spectra and uncertainty checks shown in the main text.
 - Supplementary Table S26: jshis_source_level_recompute_closure_audit.md, jshis_source_level_recompute_closure_audit.csv, and jshis_source_level_recompute_claim_boundary.csv. These files document public source-parameter checks, sampled-grid consistency tests, and the official-production claim boundary.
 - Supplementary Table S27: jshis_sa3_physical_proxy_model.md, jshis_sa3_physical_proxy_model_metrics.csv, jshis_sa3_physical_proxy_model_predictions.csv, and jshis_sa3_physical_proxy_model_coefficients.csv. These files report the interpretable site-variable model for MF2013 site-corrected SA(3.0 s) station residuals.